

## **Political Prisoner's Dilemma: Evidence from the Great Terror in the Soviet Union<sup>1</sup>**

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### **Classic prisoner's dilemma model**

The classic prisoner's dilemma predicts that it is rational for individuals, who are charged with a crime and are given a choice, to confess and implicate others in order to get lower sentences. For example, the prosecutors give each prisoner a choice of one-year sentence if he confesses and implicates another and 25 year sentence if he does not. If one prisoner confesses and implicates the other, and the other one does not confess, the person who has voluntarily helped the prosecutors will get 1 year sentence, while the other person will get 25 year prison term. If they both do not confess, the two men will get 3-year sentences even though they are both innocent. (Table 1).

[Table 1 about here]

The prisoners' dilemma model predicts that rational prisoners who are unbounded by morality and friendship will both confess and get 10 year sentence even they individually would be both better off not confessing. (Table 1). Assuming, as neoclassical economics and rational choice theory in political science do, that prisoners maximize their expected utility, confession is the optimal solution or the dominant strategy of the classic prisoner dilemma.

The logic and the outcome of the prisoner's dilemma game is essentially the same for large number of prisoners, or "players," and for other similar distribution of sentences, or pay-offs. Other solutions, involving cooperation between prisoners can arise only if they face repetitive choices, i.e. they presented with the dilemma multiple and indefinite number of times, or if external payoffs affect stakes faced by players. Tit for tat strategy beats other choices in nonfinite iterative prisoner's dilemma games.<sup>1</sup>

### **Research question and previous studies**

Many experimental tests of the prisoner's dilemma found that cooperation is a much more frequent outcome than that predicted by the model.<sup>2</sup> However, these experiments were often carried out among university students majoring in Economics. They involved artificial situations and small stakes.<sup>3</sup>

The Great Terror in the Soviet Union presents an example of the prisoner's dilemma situation.<sup>4</sup> The question is whether political prisoners during the Great Terror cooperated with prosecutors and voluntarily confessed as the prisoner's dilemma predicts.

The Great Terror in the Soviet Union came closest to the real-life prisoner's dilemma situation. It involved large numbers of individuals arrested on political charges that carried capital punishment or sentences of varying length. Several million people were arrested at the height of the terror in 1937-1938. In contrast to many previous studies which denote the start of the Great Terror with waves of political arrests that followed the assassination of Kirov in the end of 1934, this article regards political arrests in the beginning of the 1930s as part of the Great Terror. This is in large part due to the focus of the study on political prosecution of intellectuals in Ukraine which began in mass with the Union of Liberation of Ukraine (SVU) show trial in 1930. Approximately 80 percent of Ukrainian writers, 43 percent of legal scholars, 41 percent of ethnic and folk culture studies scholars and anthropologists, 37 percent of historians, 36 percent of archeologists, 33 percent of geologists, 31 percent of specialists in study of literature and language, 27 percent of economists, 25 percent of physicists, 22 percent of librarians, 20 percent of physicians, and 14 percent of mathematicians at the research centers of the All-Ukrainian Academy of Sciences were arrested in the 1930s.

Soviet prosecutors demanded from the prisoners to confess to political crimes and name associates in exchange for more lenient punishment. The confessions were used to convict arrested individuals, as well as to arrest others. Confession was a necessary and sufficient proof of guilt. Punishment for counterrevolutionary activity in the Soviet criminal code ranged from execution to 3 years or more if there were softening circumstances.<sup>5</sup> Investigators promised less severe punishment to political prisoners who cooperated with them and implicated others.

Political prisoners had to make their choices based on their expected sentences. The absolute majority of political trials were conducted in secret and the NKVD deliberately misinformed that prisoners who were executed were sentenced to 10 years. Therefore, deception by the NKVD, which often gave more severe sentences than promised, did not affect incentives of prisoners to confess.<sup>6</sup>

The totalitarian political system and the protracted sentences or executions meant that possibilities of external enforcement and repeated interaction were minimal. Trials were carried out without jury and witnesses by specially appointed courts and commissions. They overwhelmingly resulted in automatic convictions that were rarely overturned on appeal. For example, statistical data of the NKVD showed that about 200 thousands of arrested individuals in Soviet Ukraine were tried on political and spying charges in 1937-1938. In 99.7 percent of cases they were found guilty and given sentences ranging from the capital punishment (execution) to 3 years. Only 0.3 percent of cases resulted in release.<sup>7</sup>

This study combines political science approaches and experimental economics approaches and uses the quasi-experimental conditions of the Great Terror to evaluate the

prisoner's dilemma model. An increasing number of social scientists use history to test rational choice models.<sup>8</sup> However, many of these studies employ selective historical evidence and often disregard historical facts that contradict rational choice theories.<sup>9</sup> For example, one study maintains that Stalin primarily relied on economic incentives and his behavior was akin to behavior of kings and other authoritarian leaders and it was primarily motivated by maximization of state revenues.<sup>10</sup> This view disregards the role of communist ideology in motivating the Soviet ruler, who imprisoned, exiled and killed millions of people because they were regarded as class enemies. Similarly, it ignores the widespread use of imprisonment and other forms of punishment for work-related violations.

Researchers found different patterns of behavior of political prisoners during the Great Terror in the Soviet Union. For example, a previous study of the prisoner's dilemma model of the Great Terror concludes that most political prisoners confessed and implicated others as the prisoner's dilemma model predicts. Many historians emphasize mass character of both voluntary denunciations and torture during the Great Terror but they do not quantify the relative roles of these factors.<sup>11</sup>

These studies mostly rely on anecdotal accounts because archival materials on the Great Terror were closed until Gorbachev's policy of *glasnost* and the collapse of the Soviet Union in the end of the 1980s and beginning of the 1990s. Recently declassified documents from the Soviet archives show that one of the most cited publications on the role of confessions during the Great Terror was written by a secret informer of the NKVD. Because Konstantin Shteppa tried to conceal this fact he provided a distorted account of his own behavior and behavior of other political prisoners, including those examined in

this study, and emphasized voluntary denunciations and voluntary cooperation of prisoners with the NKVD.<sup>12</sup>

### **Methodology and data**

This article analyzes behavior of more than 100 political prisoners arrested in the Soviet Union in the 1930s.<sup>13</sup> These cases are linked to the political persecution of Mykhailo Krawtchouk (Kravchuk), an internationally renowned Ukrainian mathematician, who was arrested in 1938. Mykhailo Krawtchouk was an academician of the All-Ukrainian Academy of Sciences (VUAN). He held mathematics chair at the National Technical University of Ukraine, headed a mathematical statistics division in the Institute of Mathematics of the VUAN, and taught at several universities in Kyiv. Krawtchouk's contributions to different areas of mathematics, mathematical physics and statistics are still widely used.<sup>14</sup>

This article analyzes behavior of Mykhailo Krawtchouk, imprisoned individuals mentioned in the materials of his prosecution and prisoners connected to his academic and personal networks. They include people arrested on various political charges in the All-Ukrainian Academy of Sciences (VUAN), the Institute of Mathematics of the VUAN, The Institute of Physics of the VUAN, National Economic University in Kyiv, Ukrainian Physico-Technological Institute (UFTI) in Kharkiv, and the Jet Propulsion Research Institute (RNII) in Moscow, and in several other organizations. Since each of these prisoners was prosecuted as a part of a larger case, behavior of all prisoners in these cases is examined. The prosecution of more than 100 of individuals in these cases presents a large N prisoner's dilemma with several sub-dilemmas.

The Krawtchouk's case is selected because it involves externalities (unintended effects) pertaining to the invention and patenting of the electronic computer in the US and rocket technology development in the Soviet Union. Krawtchouk's mathematical work and his political persecutions had an unintended effect on the invention and patenting of the Atanasoff-Berry computer which was designed to speed up solution of mathematical equations with help of methods developed by Mykhailo Krawtchouk. Krawtchouk's students and acquaintances included prosecuted scientists and engineers, such as Sergei Korolev and Yuri Kondratyuk, who played significant roles in the masterminding the Soviet space program and developing theory of space flight.<sup>15</sup>

The data for this study are obtained from recently declassified materials of prosecution and rehabilitation of political prisoners in the former Soviet Union. This article employs archival materials of interrogations and trials in the 1930s, legal reviews and judicial rehabilitation of political prisoners in the 1950s-1990s, and secondary publications.

The same criteria are applied to evaluate each prosecution case. Confessions are classified as voluntary when prisoners during their first interrogations admitted their alleged political crimes, implicated others and did not change their testimony during subsequent interrogations, trial and in the aftermath of the trial. When archival-investigative cases or subsequent judicial reviews of individual prosecutions contain both confessions and references to use in these particular cases of interrogation methods, which produced significant physical and psychological pain, such confessions are treated as involuntary. When the testimony of prisoners radically changed after initial interrogations or during their trial, such confessions are also regarded as involuntary because torture was

the most plausible reason for such sudden confessions. When prisoners consistently refused to confess and implicate others, such behavior is classified as non confession.

### **The Krawtchouk case**

Mykhailo Krawtchouk was arrested on February 21, 1938. He was charged with a membership in an underground Ukrainian nationalist and terrorist organization and spying. The accusation of a membership in such fictitious organizations was a typical charge against intellectuals during the Great Terror. The allegations of stemmed from the fact that Krawtchouk closely identified with Ukrainian culture, playing a major role in developing Ukrainian mathematical terminology and mathematical education. Mykhailo Krawtchouk published his works in French, German, Italian, and Russian, but he wrote most of his books and articles in Ukrainian. It did not help that Krawtchouk had relatives and students from a variety of ethnic and religious backgrounds: not only Ukrainian, but also German, Jewish, Polish and Russian. He pointed out this fact to his interrogators but it had no effect on their demands to confess in the political crimes.

The NKVD prosecutors wanted Mykhailo Krawtchouk to confess about his subversive activity and to implicate others, e.g., his coworkers, students, and acquaintances in the membership in the fictitious underground organization. Krawtchouk wrote in his appeal to the Supreme Procurator of the USSR in 1940:

They demanded from me, first of all, to give a written confession of being a member of this organization. They also demanded from me to name the upper leadership of this organization and persons who recruited me and whom I recruited, and who instructed me and whom I instructed on the matters of the organization within the Academy of Sciences. They demanded to describe the evolution of my counter-revolutionary convictions, my counter-revolutionary work, participation in the White Armies, my counter-revolutionary agitation among students, creation of insurgent detachments by the members of the organization, my spying activity, etc.<sup>16</sup>



Mykhailo Krawtchouk was charged with being “an active member and a leader” of an organization of former *Borotbisty*. *Borotbisty* was a party established in 1918 by a left wing of the Ukrainian Party of Socialist Revolutionaries, which advocated a socialist and populist program, as well as the independence of Ukraine.<sup>17</sup> Mykhailo Hrushevsky, the first president of Ukraine during its brief independence in 1918 and the leading Ukrainian historian, was politically close to the party of Ukrainian socialist revolutionaries. In contrast to most socialist revolutionaries, *Borotbisty* leaders advocated cooperation with the Bolsheviks. They joined the Bolshevik party in 1920 when it prohibited the existence of all other political parties.

Mykhailo Krawtchouk was never a member of *Borotbisty* or any other party. But evidence such as his official contacts with some of the former *Borotbisty* and Hrushevsky’s history book that the NKVD confiscated during Krawtchouk’s arrest were sufficient to link the Ukrainian mathematician to the prosecution of the fictitious organization of former *Borotbisty*.

The materials of his prosecution show that Mykhailo Krawtchouk did not confess and did not implicate other individuals voluntarily. Despite demands of a NKVD interrogator, he refused to implicate former leaders of *Borotbisty* whom he knew in their capacity as high-ranking government officials. The NKVD prosecutors tried to link Krawtchouk to Andrij Khvyliia, Panas Lubchenko and Volodymyr Zatonsky. These alleged leaders of the underground organization were charged in 1937 after the NKVD obtained incriminatory testimonies during interrogation of other political prisoners.<sup>18</sup>

Krawtchouk was charged with establishing connection with the counterrevolutionary organization in 1933. At that time, Andrij Khvylya was deputy minister of education in Soviet Ukraine. When he was arrested on August 13, 1937, Khvylya was the head of the Directorate for the Arts. In his official positions and earlier as the head of the Communist party propaganda department, Khvylya publicly accused many intellectuals of political deviations. Most of these intellectuals were arrested on political charges. In one of the most notable examples, Khvylya denounced Mykola Khvylovy, a Ukrainian writer, for suggesting the slogan, "Away from Moscow." Khvylovy committed suicide in 1933 because of the public campaign against him and arrests of other Ukrainian intellectuals. In turn, Khvylya was denounced by some of his colleagues and arrested.<sup>19</sup>

During his interrogation, Andrij Khvylya confessed to being a leader, along with more than dozen of other top government officials, of the underground organization. On August 18, 1937, Khvylya wrote a declaration addressed to Stalin and other top party and NKVD officials. The declaration presented cultural and academic contacts with Western Ukrainian intellectuals in Poland as communication with foreign spies.<sup>20</sup> However, during his interrogation and confrontation with Panas Lubchenko in Moscow a few days later, Khvylya renounced his confession. Khvylya told Stalin that he and other alleged members of the organization were not guilty. Stalin, along with other members of the Soviet Communist party leadership, was present during this interrogation and confrontation. Later the same day, Khvylya changed his testimony and repeated his earlier confessions. However, during his closed trial on February 5, 1938, he again renounced his confessions. Such reversals indicated that the NKVD interrogators used torture to extract confessions from Khvylya.<sup>21</sup>

Panas Lubchenko was another alleged leader of the organization to which Krawtchouk was linked by the NKVD. In fact, Lubchenko was the head of the government of Soviet Ukraine and a member of the Political Bureau, the governing body of the Ukrainian branch of the Soviet Communist party. However, his membership in *Borotbisty* party twenty years earlier as well as confessions obtained from political prisoners led to his dismissal from the top party position. He faced an inevitable arrest. In his various prior capacities, Lubchenko took part in denouncing many government officials and Ukrainian intellectuals. For example, he served as the main public prosecutor during the 1930 show trial of 45 intellectuals falsely accused in membership in SVU (Union for Liberation of Ukraine). He did the same during a meeting of the Central Committee of the Ukrainian branch of the Soviet Communist Party in the end of August 1937. This meeting reviewed incriminating testimonies against Lubchenko that the NKVD had obtained from interrogating prisoners such as Khvyliia. Lubchenko denounced Khvyliia and other members of the alleged underground organization, but he denied his own involvement.<sup>22</sup>

Volodymyr Zatonsky was arrested on November 3, 1937. He was the Minister of Education of Soviet Ukraine, a member of Communist party leadership, and an Academician of the VUAN. He studied and then taught physics at Kyiv University around the same time that Krawtchouk did in the 1910s. Zatonsky later became one of the founders of the Ukrainian branch of the Bolshevik Party. Like many other party officials, he was involved in political denunciations, and was denounced himself during the Great Terror.<sup>23</sup>

During his interrogation by the NKVD, Zatonsky mentioned Krawtchouk's name, along with many other names, as a member of the underground organization. Materials of

his archival-investigative case, in particular, short length of time between his arrest and detailed confession, as well as the large number of individuals named by Zatonsky, indicate a voluntary nature of his confession.<sup>24</sup>

During Krawtchouk's interrogation, NKVD prosecutors tried to present his meetings with Zatonsky as contact between members of the underground organization. A protocol of Krawtchouk's interrogation stated that he informed Zatonsky about "his anti-Soviet nationalist work at the Institute of Mathematics" and received new tasks. These tasks included "promoting young nationalist cadres on the educational front," "supporting the old reactionary professorate," and "directing it against young Soviet scientists."<sup>25</sup> In fact, Mykhailo Krawtchouk visited Volodymyr Zatonsky several times in 1937 to counter accusations that he was giving low grades to students at the Kyiv Polytechnic Institute (now National Technical University of Ukraine). Some of his students and colleagues denounced Krawtchouk's teaching despite numerous testimonies to the contrary. Poor quality of work was considered sabotage, a political crime, in Stalin's Soviet Union.

Because Krawtchouk did not cooperate voluntarily with the prosecution and refused to confess and implicate others, the NKVD interrogators switched their method to torture. In his appeal letter, Krawtchouk summarized methods used by the NKVD investigators in his case as follows:

I was devastated by these wild accusations, physically broken by the night interrogations, in particular, by complete sleep deprivation for 11 days and nights, worsening of my heart disease, and means of direct physical force. They were coercing me morally by screams and moans of the people being tortured in the neighboring rooms.<sup>26</sup>

The NKVD frequently used such methods of interrogation to force political prisoners to confess.<sup>27</sup> In addition to beatings and psychological intimidation, Krawtchouk

described a torture method called “conveyer.” It involved constant interrogation of political prisoners by NKVD interrogators and deliberate sleep deprivation that could last up to several weeks. While prisoners were not allowed to rest or sleep, the interrogators rotated daily.

By means of torture in prison and by threats to arrest his family, prosecutors forced Krawtchouk to sign confession statements. Before his trial, new NKVD interrogators threatened Krawtchouk not to retract his confession because he would remain under control of the Soviet organs.<sup>28</sup> However, Krawtchouk did retract his confession after the trial ended.

The behavior of Mykhailo Krawtchouk during his interrogations contradicts the prisoner’s dilemma model. He did not confess voluntarily in exchange for a more lenient punishment. His sentence reaffirmed the high stakes involved in his choice not to confess voluntarily. On September 23, 1938, in a trial lasting one half-hour, the Military Collegium of the Supreme Court of the USSR sentenced Krawtchouk to 20 years in prison and 5 years in exile. Mykhailo Krawtchouk was assigned to perform heavy manual work in a gold mine in Kolyma region. Like a large proportion of political prisoners, Krawtchouk did not survive the Kolyma camps. He died there in March 1942.

Mykhailo Krawtchouk was convicted for being a member of the Ukrainian nationalist and terrorist organization, which “included” three of his former students. Volodymyr Mozhar, a mathematics professor at the Kyiv Technologic Institute of Food Production (now National University of Food Technologies), was one of them. He was arrested on May 1, 1937, and charged with being an active member of the Ukrainian nationalist terrorist organization.<sup>29</sup> Mozhar was a researcher at the Division of

Mathematical Statistics at the Institute of Mathematics. Krawtchouk headed this division. They also co-authored two books on mathematics.<sup>30</sup>

Mykhailo Krawtchouk refused to describe Mozhar as an “enemy of the people” in an academic article he published in 1937.<sup>31</sup> This fact was used to initiate a public campaign of Krawtchouk’s denunciation that led to his dismissal from the academic positions and the eventual arrest. *Kommunist*, the main official newspaper in Soviet Ukraine, published on September 14, 1937, an editorial titled “Academician Krawtchouk Advertises Enemies.” The editorial was signed by Dmytro Grave, the director of the Institute of Mathematics of the Ukrainian Academy of Sciences, and Kindrat Breus, the academic secretary of the Institute. The article stated that in his piece in the journal *Uspekhi Matematicheskikh Nauk* (*Progress in Mathematical Sciences*), Krawtchouk gave a favorable review of the work of “enemies of the people” who were uncovered by the NKVD. The letter noted that Mykhailo Krawtchouk had enough time to withdraw his article or to make “appropriate corrections,” but he chose not to.

Volodymyr Mozhar did not confess during his interrogation in his membership in the underground organization and he did not implicate Krawtchouk and others. His first interrogation after being transported from Moscow, where he was arrested, to a prison in Kyiv, is dated May 10, 1937. The protocol of his interrogation stated that Mozhar categorically denied his membership in the organization. He was specifically asked by Gaponov, a NKVD interrogator, to implicate Mykhailo Krawtchouk: “You are a student of Krawtchouk, a known nationalist.” Mozhar confirmed that he was Krawtchouk’s student but said that he did not know about the nationalist views of his teacher.<sup>32</sup>

Volodymyr Mozhar's membership in 1919 in the *Borotbisty party* as well as books found during his arrest served as evidence for nationalist charges. NKVD documents characterized books confiscated during Mozhar's arrest as "counterrevolutionary and ideologically dangerous." Almost all these books were published legally in the Soviet Union by intellectuals who were arrested during the Great Terror. The NKVD also used Mozhar's earlier contacts with several anarchists and a confiscated book by Peter Kropotkin, a famous 19<sup>th</sup> century Russian anarchist, to accuse him of membership in an underground anarchist organization. It relied on testimony obtained from several other individuals to corroborate these charges. During his interrogation on June 22, 1937, a NKVD interrogator told Mozhar that the investigation had detailed evidence about his "participation in a counterrevolutionary organization" and persons with whom he "jointly conducted counterrevolutionary activity." Mozhar again refused to confess in the membership in the underground organization.<sup>33</sup>

Testimonies of three other imprisoned individuals and one witness provided the NKVD with incriminatory evidence against Mozhar. Confessions by two of these political prisoners were not voluntary. Circumstances of the confession of the third prisoner cannot be established because archival materials and publications about Mozhar's prosecution include incomplete materials regarding this prisoner, who was arrested in Leningrad (now St. Petersburg) in April 1937.<sup>34</sup>

The confession of Israel Dubinsky, who was arrested on June 12, 1937, and whose case had the same number as Mozhar's case, was obtained by the NKVD by means of torture. In 1956, Dubinsky testified to a KGB investigator who was reviewing his case that all his confessions "furnished in 1937 were given as result of unbearable physical

coercion” and that investigators even were throwing a cat on his head and on his face.<sup>35</sup> In 1939, Dubinsky lodged a complaint with the Prosecutor General of the USSR about his conviction. He wrote that the NKVD interrogator distorted his testimony, in particular, by presenting a gathering at Mozhar’s apartment as a meeting of counterrevolutionary organization.<sup>36</sup>

The prosecution case of Mozhar and Dubinsky contains testimony of Bentsion Aisenberg, who was arrested in April of 1937 on charges of membership in an anarchist-Trotskyist terrorist organization. In his letter to the head of the NKVD in August 1937, Aisenberg named Mozhar and Dubinsky as members of the same organization, but he also stated that he abandoned his earlier refusal to confess.<sup>37</sup> Such sudden change in behavior is an indication that his confession was not voluntary.

Mozhar refused to confess during the interrogation and trial to all charges except “anti-Soviet agitation” and was given a capital punishment. The refusal to confess was contrary to his self-interest, as the prisoner’s dilemma shows. Mozhar paid the ultimate price for non-cooperation with the NKVD, even though he had the chance to incriminate Mykhailo Krawtchouk and others. Mozhar was executed immediately after the verdict was announced on the 9<sup>th</sup> of November in 1937. As in many other cases, the verdict was not made public. Only in the end of the 1980s, the family of Mozhar learned that he was executed. Israel Dubinsky, who signed a confession under torture, was sentenced to 10 years.<sup>38</sup> The behavior of both prisoners in this case contradicts the prisoner’s dilemma model. Despite incentives to confess offered by the NKVD, neither prisoner chose to cooperate with the prosecutors voluntarily and implicate others.



David Topolyansky, a mathematics professor and Krawtchouk's colleague at Kyiv University and the Division of Mathematical Statistics at the Institute of Mathematics, was arrested on June 11, 1937. Krawtchouk and Topolyansky co-authored an article in a mathematical journal. In another article published less than a year later, Krawtchouk mentioned his joint work with Topolyansky at the Institute of Mathematics of the VUAN. As in the case of Mozhar, this fact was held against Krawtchouk during the public campaign against him which began in September 1937. During his interrogation and trial, Krawtchouk did not implicate Topolyansky in membership in any fictitious underground organizations.

Similarly, David Topolyansky refused to cooperate with the prosecutors and implicate Mykhailo Krawtchouk and others. A summary statement of the prosecution in October of 1938 concluded that "the investigation conducted in this case established that Topolyansky, while being a member of anti-Soviet Trotskyist organization, was connected with Sachnovsky and Krawtchouk (sentenced), members of this organization, and conducted anti-Soviet activity."<sup>39</sup> However, the testimony about Krawtchouk came not from Topolyansky, but from two witnesses. These witnesses, who were graduate students at Kyiv University, informed the NKVD that Krawtchouk and Sachnovsky, the rector of the university, defended Topolyansky against accusations of poor teaching quality and helped to promote him to the position of a professor.<sup>40</sup> A summary statement of the prosecution accused Topolyansky of using "bourgeois methods in teaching in order to harm communist education of the youth." According to the NKVD indictment, Topolyansky said that only talented people could quickly learn mathematics.<sup>41</sup>

David Topolyansky was released in the end of December, 1938, after about a year and a half in prison, due to the lack of evidence against him. His refusal to confess was a major factor in his release. Another major factor, though, was that Krawtchouk and Mozhar did not implicate him during their interrogations, even though it was in their self-interests, as the prisoner's dilemma model shows. The NKVD documents stated that testimony obtained from Sachnovsky and used to arrest Topolyansky was not incriminating enough to convict him. The testimony of Grygory Lozovyk, a professor of ancient history at Kyiv University, against Topolyansky was discounted because it relied on words of Sachnovsky.<sup>42</sup> This was not for lack of effort by the NKVD. Sachnovsky and Lozovyk, who were arrested as a part of the fictitious "Trotskyist Terrorist Organization" in the beginning of 1936, were already sentenced and executed.<sup>43</sup> Two witnesses in the case did not provide enough evidence about "anti-Soviet activity" of Topolyansky.<sup>44</sup>

During his interrogation, Mykhailo Krawtchouk refused to present his academic contacts with foreign mathematicians as spying and to implicate other political prisoners in this fictitious crime, although the prisoners' dilemma predicts that this went against his self-interest. NKVD interrogators queried Mykhailo Krawtchouk about his foreign ties in order to portray these contacts as spying activity. Krawtchouk described his contacts with West Ukrainian, French, and German mathematicians and physicists, but he refused to confess in spying.<sup>45</sup> Krawtchouk stated that he suspended his foreign correspondence and publication of his papers abroad in 1936 to avoid any repercussions. Before 1936, Krawtchouk maintained academic and personal contacts with many noted foreign mathematicians. They included Jacques Hadamard in France, David Hilbert and Richard Courant in Germany, and Tullio Levi-Civita and Francesco Tricomi in Italy. Mykhailo

Krawtchouk received job offers from several leading US universities in the 1920s, but he declined them.

The NKVD charged Krawtchouk with helping Mykola Tchaikovsky to spy for Poland. Tchaikovsky, a Ukrainian mathematician, was born in the Galicia province, then part of Austria, and was educated at the University of Vienna. After World War I, Galicia became part of Poland and the Polish authorities restricted Ukrainian-language education in universities and schools. Krawtchouk helped Tchaikovsky immigrate to Soviet Ukraine and found him an academic position in Odessa University. The NKVD used this fact as evidence for spying charges against Krawtchouk. Tchaikovsky was arrested in 1933, and he was falsely charged with being a spy and a member of the underground Ukrainian Military Organization (UVO). The NKVD interrogators presented the fact that he once lived in Poland and visited France as evidence that he was a spy and asked him to choose whether he wanted to confess of being a Polish spy or a French spy. Tchaikovsky did not confess voluntarily during his interrogation.<sup>46</sup>

Although Mykhailo Krawtchouk was accused with ties to Borys Lagunov, he did not provide any incriminating evidence against Lagunov. They became acquainted during Lagunov's brief work at the Institute of Mathematics in the beginning of the 1930s. Borys Lagunov was arrested along with 11 other individuals in 1936. The NKVD charged him with active participation in an underground organization "People Party" and with preparation of terrorist attacks against Soviet party and government leaders. However, Lagunov and other prisoners, with one exception, refused to confess voluntarily during their interrogation, and they were tortured. He and other prisoners in his case were sentenced to 3-5 years of imprisonment or exile. Borys Lagunov was arrested again on the

same charges in a GULAG camp in January 1938. However, materials about his second arrest are insufficient to determine whether he confessed voluntarily or involuntarily.<sup>47</sup>

Twelve political prisoners implicated Krawtchouk in the membership in the underground organizations and other political crimes, but they did this involuntarily, i.e. as result of physical and psychological torture during their interrogations.<sup>48</sup> These prisoners included Mykhailo Ptukha, Director of the Institute of Demography of the VUAN, Volodymyr Ivanushkin, Director of the Library of the VUAN, Fedir Polonsky, a researcher at the Institute of Geology of the VUAN, Maxim Lukovsky, former secretary of the party committee of the VUAN, Leonid Levytsky, Deputy Director of the Institute of Experimental Biology of the VUAN, Josip Hermaize, history professor at Kyiv University, Volodymyr Durdukovsky, director of a school in Kyiv, Grygori Kholodny, Director of the Institute of Ukrainian Scientific Language of the VUAN, Grygori Ivanytsya, professor of Ukrainian language at Kyiv University, Petro Suprunenko, Director of the Institute of Transportation Mechanics of the VUAN, and Arnold Krister, professor of law at the National Economic University in Kyiv.<sup>49</sup>

They all refused to confess voluntarily and were tortured. The physical and psychological torture methods used by the NKVD in these and other similar cases included sleep deprivation and protracted interrogations up to several weeks, beating, forced isolation, food deprivation, forced standing in stress positions, threats of immediate execution of the prisoner and his family and forced witnessing of torture of other prisoners.

### **Defection**

Some political prisoners “defected” and voluntarily denounced Krawtchouk and other individuals. For example, Stepan Feshchenko, a researcher at the Institute of

Mathematics and a Communist party candidate, who was arrested in December of 1937, voluntarily denounced Mykhailo Krawtchouk during his interrogation. Feshchenko disliked Krawtchouk, who thought him lacking in mathematical knowledge and ability. A NKVD operative testified that Feshchenko wrote a letter to the NKVD in the beginning of 1936 informing about the “anti-Soviet feelings and nationalist views” of Academician Krawtchouk.<sup>50</sup>

Stepan Feshchenko stated that Mozhar and Topolyansky were arrested because of his denunciation to the NKVD. He also said that Mykhailo Orlov, the academic secretary of the Institute of Mathematics and his friend, was arrested in 1936 because of his denunciation. Feshchenko declared that he wanted to denounce others. He was released on February 22, 1938, as a reward for his denunciations.<sup>51</sup>

Mykhailo Orlov was charged in the membership in the Trotskyist Terrorist Organization. In addition to his position at the Institute of Mathematics, Orlov was a high-ranking Communist party official. Archival materials of his prosecution indicate that he confessed voluntarily. Three days after his arrest, Orlov admitted his membership in this fictitious organization and implicated others.<sup>52</sup>

As a *seksot* (secret informer) of the NKVD since 1935, Serhii Tutkovsky (Lvovych) provided incrimination and false information that led to arrests and convictions of Volodymyr Mozhar, Borys Lagunov and many other people connected to Mykhailo Krawtchouk’s case. Tutkovsky was a former *Borotbist*. He joined the Bolshevik party and taught philosophy at Kyiv University. Tutkovsky confessed voluntarily soon after his arrest in September 1937 in a membership in the fictitious Trotskyist Terrorist Organization, which he helped to create as a secret informer of the NKVD.<sup>53</sup>

Konstantyn Shtepa, professor of ancient history at Kyiv University, was also involved in Krawtchouk's case and voluntarily cooperated with the NKVD. He was arrested in March of 1938 and charged with participation in an underground Ukrainian nationalist organization and spying for Japan. However, archival documents and research by historians show that Konstantyn Shtepa was a *seksot* who denounced people to the NKVD.<sup>54</sup> At the special session of the military tribunal on April 16, 1939, Shtepa stated that the NKVD gave him especially important tasks in uncovering Ukrainian nationalist organizations and that he provided much material to accomplish these tasks. He also claimed credit for uncovering "the nationalist group" headed by Sachnovsky, Lozovyk, and others at Kyiv University.<sup>55</sup>

Reports of secret informers to the NKVD in Krawtchouk's case are not yet opened for researchers. However, indirect evidence points towards Konstantyn Shtepa as one of the secret informers in this and other related cases. In the mid 1930s, Shtepa established ties with Krawtchouk, even though they worked in very different academic fields. As Shtepa himself acknowledged during the closed session of the trial, the NKVD officials installed secret informers close to people they considered politically suspicious.<sup>56</sup>

Shtepa's behavior during the Great Terror fits the prisoner's dilemma model. He is an example of a rational self-interested individual, unconstrained by any ethical values. Throughout his life, Konstantyn Shtepa constantly changed his behavior and political stance according to changes in incentives. He fought the Bolsheviks during the Civil War, but turned into "a non-party Bolshevik" after they won the war. Archival documents recently uncovered by historians show that during the Nazi occupation of Ukraine, Shtepa became a German secret police informer. He became the chief editor of the main pro-

German newspaper in Kyiv in the end of 1941, after the Nazis executed the paper's previous editors for their association with the Organization of Ukrainian Nationalists. He published many pro-Nazi, anti-Soviet, anti-Semitic and anti-American articles, and earned praise for his active collaboration from the commander-in-chief of the German Army stationed in Ukraine.<sup>57</sup> However, after the war ended with the Nazi defeat, Shteppa again conveniently changed his behavior and public views. For instance, he received an immigration sponsorship and material assistance from Isaac Don Levine, a Jewish immigrant from the Russian Empire and a prominent American journalist. Similarly, Shteppa used support from Niels Bohr and Alexander Dallin, Jewish scholars, in publishing his books.<sup>58</sup>

In his books, Konstantyn Shteppa portrayed himself as a victim of the Great Terror. He described several secret informers but failed to acknowledge his service to the NKVD.<sup>59</sup> Contrary to his claims in these publications, Shteppa provided distorted accounts of his own behavior and behavior of other people involved in his case. For example, Konstantyn Shteppa referred in his publications to Georgy Lozovyk, a history professor at Kyiv University, as his close friend.<sup>60</sup> But, as noted, Shteppa voluntarily stated during his trial that he denounced Lozovyk to the NKVD. Georgy Lozovyk was arrested as a former Menshevik. After he left the Soviet Union, Konstantyn Shteppa established close ties with Borys Nikolaevsky, one of the leaders of émigré Mensheviks in the US. Shteppa received assistance from Nikolaevsky in obtaining a Ford Foundation grant for his book on Soviet historians.<sup>61</sup>

### **The Union of Liberation of Ukraine (SVU) Case**

Krawtchouk's name was mentioned in several testimonies obtained by the NKVD prosecutors from the accused in a big show trial held in 1930. Forty five intellectuals were accused in forming an underground organization, the Union of Liberation of Ukraine (SVU). The fictitious accusations and the show trial were masterminded by the Communist party leaders and the GPU, as the NKVD was then called. Stalin himself sent a secret letter to Kossior and Chubar, Communist Party leaders of Ukraine, with instructions on how to stage the show trial and add new charges of poisoning communist leadership. Mykhailo Krawtchouk was appointed as a public prosecutor at the SVU show trial but he declined to participate despite significant risks that this decision entailed. During the public campaign against Krawtchouk in 1937, this fact was used as an evidence of his anti-Soviet activity.

Materials of Krawtchouk's prosecution state that Josip Hermaize recruited him in 1928 into a "counter-revolutionary and terrorist organization." Josip Hermaize, a Jewish professor of Ukrainian history at Kyiv University and head of the Archeographic commission of the VUAN, was convicted as one of the leaders of the SVU in 1930. Krawtchouk and Hermaize studied together at Kyiv University in the 1910s. They taught at the First Ukrainian Gymnasium in 1919-1920 and then at Kyiv University.

During his interrogation at the SVU case, Josip Hermaize like the majority of other prisoners, refused to give voluntary confession for several months. In 1989, a review of the materials of the SVU case by a special commission concluded that 42 out of 45 accused did not confess voluntarily during their interrogations.<sup>62</sup> However, Hermaize and most of the others were forced by the NKVD to change their testimony and implicate each other and Mykhailo Krawtchouk.<sup>63</sup>



## **Interrogation of Rocket Scientists**

Serhii Korolev, a former Krawtchouk's student at the Kyiv Polytechnic Institute, was arrested in Moscow in June 1938. Mykhailo Krawtchouk helped Korolev to get admission to the Kyiv Polytechnic Institute by waiving a two-year work experience requirement introduced for people of non-proletarian origin.<sup>64</sup> Korolev later transferred to a Moscow Institute and joined in the 1930s a rocket program headed by Tukhachevsky, a Red Army Marshall.

Serhii Korolev was arrested after his imprisoned colleagues at the rocketry institute provided testimony which implicated him. However, this testimony was not voluntary.<sup>65</sup> Korolev refused to confess voluntarily and was tortured. He confessed but later renounced his testimony. Korolev was implicated by testimonies of Langemak and Kleimenov, two leaders of the Jet Propulsion Institute (RNII).<sup>66</sup> Their testimony was obtained by means of torture. Georgy Langemak and Ivan Kleimenov were arrested on November 2, 1937 following arrest of Tukhachevsky and several other top Soviet military officials who were falsely accused of being German agents.<sup>67</sup> Valentyn Glushko, arrested in March 1938, also confessed of being a member of the anti-Soviet organization and a German spy. However, he renounced his testimony in letters to Stalin and Andriy Vyshinsky, the prosecutor-general, and stated that he was tortured during his interrogation.<sup>68</sup>

The behavior of these four prisoners contradicts the prisoner's dilemma model. During their interrogations and trials they did not confess voluntarily in exchange for a more lenient punishment. Langemak and Kleimenov were executed. Korolev received 10 year sentence, while Glushko was given 8 years. The difference in length of sentences reflected primarily differences in their alleged roles in underground organizations.

This case had externalities pertaining to the Soviet space program. In the 1950s- and 1960s, Serhii Korolev was Chief Designer of the Soviet space program, and he masterminded launches of the first artificial satellite (*Sputnik*), the first man and first woman in space, and the first spacecrafts to reach the Moon and Venus and fly by Mars.<sup>69</sup> Korolev's belief that he was prosecuted because of Glushko's testimony led to his tensions with Glushko who became a leading rocket engine designer. This had a negative effect on the Soviet manned flight program to the Moon in the 1960s, and is considered one of reasons for the failure of this program. Korolev chose a much less experienced jet engine designer instead of Glushko to develop rocket engines for a large rocket which was supposed to fly Soviet cosmonauts to the Moon. All launches of this rocket ended in failure because of problems with engines. Unexpected death of Korolev during a surgery in 1966, which was in part attributed to a broken skull during his interrogation by the NKVD, also negatively affected the Soviet space efforts, including the secret program of the manned flight to the Moon.<sup>70</sup>

NKVD interrogators questioned Mykhailo Krawtchouk about his acquaintances from the Volyn region, in particular about family of Kondratyuks. Oleksander Shargei, who is now regarded as one of the founders of the theory of space flight, adopted documents and identity of Yuri Kondratyuk, a member of this family. Shargei did this to hide his involuntary service in the White Army from the Soviet authorities. Such service and adoption of fake identity documents were regarded as a political crime and carried a capital punishment in the Soviet Union.<sup>71</sup>

Mykhailo Krawtchouk was probably the only person who personally knew both real Yuri Kondratyuk and Oleksander Shargei. Shargei used identity of Yuri Kondratyuk

when he visited Krawtchouk in 1933 for consultations concerning mathematical calculations for a design of a large wind-powered electricity station in Crimea. This station was not built because of political arrests in a Soviet ministry which funded this project. However, a number of Shargei's ideas were later adapted by his collaborator in designing Ostankino TV tower in Moscow. Erected in 1967, this was the tallest building in the world.<sup>72</sup>

Mykhailo Krawtchouk did not reveal anything that might have led to the uncovering of this illegal identity change by Oleksander Shargei. He also did not implicate Volodymyr Kondratyuk, Yuri Kondratyuk's brother. Volodymyr Kondratyuk provided Shargei with the documents of his brother who died in 1921. If Krawtchouk were to implicate Volodymyr Kondratyuk or Oleksander Shargei, there was a significant possibility that their secret would have been uncovered by the NKVD. However, Krawtchouk minimized his contacts with Kondratyuks during his interrogation and stated that they were people of Russian culture, even though Volodymyr Kondratyuk was an ethnic Ukrainian and worked as a teacher of Ukrainian literature in the 1920s.<sup>73</sup>

Similarly, Volodymyr Kondratyuk did not confess during his interrogations about the fact that he provided Shargei with identity documents of his brother. Volodymyr Kondratyuk was first arrested in 1930 along with four other teachers of a high school in Kyiv. He and other teachers were accused in membership in a fictitious counterrevolutionary organization. Volodymyr Kondratyuk did not implicate his fellow teachers and Mykhailo Krawtchouk. Because the NKVD prosecutors could not obtain confessions from the arrested individuals, Volodymyr Kondratyuk was released after spending 6 months in prison.<sup>74</sup> In 1932, he was arrested again on similar charges.

However, Volodymyr Kondratyuk again did not confess. He was released after three months of interrogation.<sup>75</sup>

Oleksander Shargei, who lived under name of Yuri Kondratyuk, was arrested in Siberia in 1930 on charges of “sabotage” along with several other engineers. Materials of his prosecution show that he did not confess and implicate others during his interrogation. The uncovering of real identity of Shargei and likely disappearance of his works from the public domain could have affected not only his fate but also the competition between the Soviet and US space programs, in particular, their efforts in the manned flight to the Moon. Shargei-Kondratyuk was the first to suggest a lunar orbiter rendezvous (LOR) method of flying to the Moon. He proposed many other fundamental solutions, which were later used by the Soviet and American space programs, such as use of atmosphere to slow down descent of spacecraft.

The LOR method, employed by the NASA during the Apollo program, had significant technological and economic advantages over the earth-orbit rendezvous method suggested by Von Braun and the direct descent method which was also considered by the NASA in the early 1960s. The LOR is credited by experts with enabling the US to land on the Moon by the end of the 1960s, a key political objective that president Kennedy declared in 1961 to counter the Soviet lead in space. This method also saved several billions of dollars compared to other methods considered by the Apollo Program.<sup>76</sup>

The arrest of Shargei-Kondratyuk in the 1920s and his disappearance during World War II made not only him but also his ideas, like the LOR, politically dangerous in the Soviet Union. After Nazi Germany attacked the Soviet Union, Kondratyuk became a soldier in the Red Army. He was either killed in action or was captured and killed by the

Nazis along several million other Soviet POWs.<sup>77</sup> Captivity was considered as treason by the Soviet government. Discovery of Kondratyuk's work in German territory captured by the Soviet troops in the end of World War II created suspicions that he might have worked for the Nazi missile program. Soviet rocket scientists initially preferred the Earth-orbit rendezvous method for their manned Moon program, and this deprived them of a key advantage compared to the US Apollo program. They switched to the LOR concept only in 1964, more than two years after the Apollo program did.<sup>78</sup>

### **The UFTI case**

In the 1930s mathematicians and physicists belonged to the same division of the VUAN, worked in the same university departments, and their research often intersected. Krawtchouk's works in mathematical physics spanned both disciplines. Contrary to the prisoner's dilemma model, Mykhailo Krawtchouk did not implicate voluntarily any physicists including Oleksander Goldman. Goldman, director of the Institute of Physics of the VUAN in Kyiv, chaired the special commission of the Academy that was set-up following the publication of the op-ed denouncing Mykhailo Krawtchouk in September 1937 and investigated allegations against him in assisting "enemies of people" and spies. The commission concluded that these charges were valid and recommended to dismiss Krawtchouk from his position as the head of a division at the Institute of Mathematics.

Oleksander Goldman was himself arrested in January 1938 on charges of spying and membership in a Ukrainian nationalist organization. These fictitious charges were based on testimonies of other prisoners, reports by secret informers, and intercepted foreign correspondence. In contrast to his actions during the work of the special

commission, Goldman during his interrogation and trial did not confess and did not implicate Krawtchouk and others.<sup>79</sup>

Sixteen people, including the noted physicists, such as Lev Landau, a future Nobel Prize winner, were arrested in the case of the Ukrainian Physico-Technological Institute (UFTI). They were charged with counterrevolutionary activity, spying for Germany, Poland and Great Britain, and sabotage. These charges stemmed from the fact that several refugee physicists from Germany and Austria worked at the Institute and that many Soviet physicists visited the Rutherford laboratory at Cambridge and maintained contacts with their foreign colleagues.

The UFTI was headed by Oleksander Leipunsky. Like Krawtchouk and Goldman, Leipunsky was one of a handful physicists and mathematicians elected as Academicians of the VUAN. Leipunsky was arrested on June 14<sup>th</sup> 1938 and charged with being a German and British agent. On June 24<sup>th</sup> he signed a confession which implicated Fritz Houtermans, Oleksander Weissberg, Lev Landau, and Lev Shubnikov in spying, but he only admitted helping these “enemies of the people.” Leipunsky was released on August 9, 1938 ostensibly because of the lack of evidence even though several other prisoners implicated him under torture.<sup>80</sup>

Lev Shubnikov, head of the laboratory of low temperatures at the UFTI, confessed in spying and other counterrevolutionary activity the day after his arrest on August 6, 1937. He implicated many other physicists from the UFTI. Shubnikov hoped that his voluntary confession would reduce his punishment and avert arrest his pregnant wife. However, he was executed. Only in the beginning of the 1990s the fact of his execution was revealed to his wife and to the public.<sup>81</sup>

The materials of their prosecution show that three physicists arrested in the UFTI case confessed voluntarily, and 10 did not confess voluntarily. Six of the prisoners, who refused to confess voluntarily, pleaded guilty because of physical and psychological torture. Three of the prisoners did not confess even under torture.<sup>82</sup>

### **Conclusion**

The prisoner's dilemma model applied to the Great Terror in the Soviet Union implies that political prisoners had an incentive to confess and implicate other prisoners. Since the other prisoners had similar incentives, they also confessed. As a result, all of the prisoners ended up worse off than if they all refused to confess.

The analysis of Krawtchouk's case and related cases, such as the Union of Liberation of Ukraine (SVU), the Jet Propulsion Institute, and UFTI cases, finds weak support for the prisoner's dilemma model. Among 177 cases of individuals arrested on political charges in these cases, 13 percent gave voluntarily confessions and implicated other prisoners. The absolute majority of the prisoners who provided voluntary confessions and implicated others were Communist party members or secret informers of the NKVD. When this factor is held constant, the rates of voluntary confessions among Ukrainians, Jews and Russians do not differ significantly.

[Table 2 about here]

The absolute majority of the political prisoners, 70 percent, did not confess voluntarily. Their interrogators had to resort to other methods in the form of physical and psychological torture, such as sleep deprivation, to extract confessions. Most of the political prisoners confessed as a result of the torture by the Soviet secret police. This

result does not prove the effectiveness of torture, because these individuals admitted crimes they never committed.



Table 1. Classic prisoner's dilemma

|            |                |                |
|------------|----------------|----------------|
| Prisoner A | Prisoner B     |                |
|            | Do not confess | Confess        |
|            | Do not confess | Confess        |
|            | (3,3)          | (25,1)         |
| Confess    | (1,25)         | <b>(10,10)</b> |

Table 2. Behavior of political prisoners in Krawtchouk's case and related cases, percent

|                             | Cases |
|-----------------------------|-------|
| Confessed voluntarily       | 13    |
| Did not confess voluntarily | 70    |
| DK/NS                       | 17    |
| Total                       | 100   |
| N                           | 177   |

## Notes

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<sup>1</sup> See Robert Axelrod, *Evolution of Cooperation* (New York, Basic Books, 1984).

<sup>2</sup> See, for example, James Andreoni and John H. Miller, 'Rational Cooperation in the Finitely Repeated Prisoner's Dilemma: Experimental Evidence,' *Economic Journal*, 103, 1993, pp. 570-585; Donald Green and Ian Shapiro, *Pathologies of Rational Choice Theory: A critique of Applications in Political Science* (New Haven, Yale University Press, 1994), pp. 88-93.

<sup>3</sup> For instance, some economists explain gambling, lottery and other empirical anomalies of the expected utility theory such as the Allais paradox by arguing that rational individuals have different degree of risk aversion and utility preferences when different levels of wealth are at stake. See Milton Friedman and L. P. Savage, 'The Expected-Utility Hypothesis and the Measurability of Utility,' *Journal of Political Economy*, 60, December, 1952, pp. 463-474; Maurice Allais, 'Le Comportement de l' Homme Rationnel devant le Risque. Critique des Postulats et Axiomes de l' École Américaine,' *Econometrica*, 21, 4, 1953, pp. 503-546.

<sup>4</sup> See Avinash Dixit and Barry Nalebuff. *Thinking strategically: the competitive edge in business, politics, and everyday life* (New York, Norton, 1991), pp. 11-12; Peter Grossman, 'The Dilemma of Prisoners: Choice During Stalin's Great Terror, 1936-1938,' *Journal of Conflict Resolution*, 38, 1, 1994.

<sup>5</sup> See the Article 58 of the Criminal Code of the Russian Federation. Similar articles and punishments existed in other Soviet Republics, including Ukraine.

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<sup>6</sup> Grossman, 'The Dilemma of Prisoners,' pp. 43-55 at p. 46.

<sup>7</sup> Calculated from G. K. Kovtun, V. A. Voynalovich and Y. Z. Danyliuk, 'Masovi nezakonni repressii 20-ch-pochatku 50-ch rokiv na Poltavshyni,' in P. T. Tronko (ed), *Reabilitovani Istorieyu* (Kyiv, Ridnyi Krai, 1992), pp. 5-49, at p. 21.

<sup>8</sup> See Robert H. Bates, Avner Greif, Margaret Levi, Jean-Laurent Rosenthal and Barry Weingast, *Analytic Narratives* (Princeton, Princeton University Press, 1998); Avner Greif, 'Cultural Beliefs and the Organization of Society: A Historical and Theoretical Reflection on Collectivist and Individualist Societies,' *Journal of Political Economy*, 102, 5, 1994, pp. 912-950; Mancur Olson, *Power and Prosperity: Outgrowing Communist and Capitalist Dictatorships*, New York, Basic Books, 2000.

<sup>9</sup> See Jon Elster, 'Rational choice history: A case of excessive ambition,' *American Political Science Review*, 94, 3, 2000, pp. 685-695.

<sup>10</sup> Olson, *Power and Prosperity*.

<sup>11</sup> See, for example, Conquest, *The Great Terror* and Robert Tucker, *Stalin in Power: The Revolution from Above 1928-1941* (New York, Norton, 1990).

<sup>12</sup> Shteppa used the pseudonym Godin in some of his publications. See F. Beck and W. Godin, *Russian Purge and the Extraction of Confession* (New York, Viking Press, 1951), Konstantin Shteppa, 'Ezhovshina,' in A.V. Popov (ed), *XX Vek--Istoriia Odnoi Semji* (Moskva, Rusaki, 2003); Ihor Verba, 'Kost Shteppa,' *Ukrainskiy Istorychnyi Zhurnal*, 3-4, 1999, pp. 97-112, 98-114.

<sup>13</sup> Some individuals were arrested twice.

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<sup>14</sup> See Nina Virchenko, Ivan Katchanovski, Viktor Haidey, Roman Andrushkiw and Roman Voronka, (Eds), *Development of the Mathematical Ideas of Mykhailo Kravchuk (Krawtchouk)* (Kyiv-New York, National Technical University of Ukraine and Shevchenko Scientific Society (USA), 2004).

<sup>15</sup> Virchenko, Katchanovski, Haidey, Andrushkiw and Voronka, *Development of the Mathematical Ideas of Mykhailo Kravchuk (Krawtchouk)*; Mykola Soroka, *Kolymaska Theorema Kravchuka* (Kyiv, Molod, 1991).

<sup>16</sup> Krawtchouk Archival Investigative Case (SBU (State Security Service of Ukraine) Archive, Kyiv, No. 38009FP), Vol. 2, p. 109.

<sup>17</sup> The party name is derived from the name of its newspaper, which means “struggle” in Ukrainian.

<sup>18</sup> Volodymyr Urbansky, *Mykhail Filipovich Kravchuk* (Moskva, Nauka, 2002), pp. 189-192, 223-256.

<sup>19</sup> Yuri Shapoval, *Ukraina 20-50-kh Rokiv: Storinky Nenapysanoi Istorii* (Kyiv, Naukova Dumka, 1993), p. 228.

<sup>20</sup> *Ibid.*, pp. 225-226.

<sup>21</sup> *Ibid.*, pp. 228-229 and Khvyliia Archival Investigative Case (SBU Archive, Kyiv, No. 38353FP).

<sup>22</sup> Shapoval, *Ukraina 20-50-kh Rokiv*, pp. 232-237.

<sup>23</sup> Zatonsky and Khvyliia, who as alleged leaders of the counterrevolutionary organization faced severest punishment, were executed in 1938. Facing an imminent arrest, Panas Lubchenko committed a suicide after a party leadership meeting that heard his case.

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<sup>24</sup> See Zatonsky Archival Investigative Case (SBU Archive, Kyiv, No. 49834FP); B. Kiyak and O. Proskura, 'Dolya Akademika Oleksandra Goldmana,' *Z Archiviv VUCHK-GPU-NKVD-KGB*, 1-2, 1997, pp. 253-262, 256-257.

<sup>25</sup> Galyna Syta, 'Stradnytskij Shlach Academica Mykhaila Kravchuka,' *Zona*, 5, 1993, pp. 100-116 at pp. 107-108.

<sup>26</sup> Galyna Syta, 'Slidcha Sprava Mykhaila Kravchuka,' *Ukraina: Nauka I Kultura*, 28, pp. 116-127; Krawtchouk Archival Investigative Case, Vol. 2, p. 175.

<sup>27</sup> See Conquest, *The Great Terror* and Alexander Weissberg, *The Accused*, translated by Edward Fitzgerald (New York, Simon and Schuster, 1951).

<sup>28</sup> The reasons for the transfer of Krawtchouk's case to new investigators are not known.

<sup>29</sup> Mozhar and Dubinsky Archival Investigative Case (TsGAOO (Central State Archive of Public Organizations of Ukraine), Kyiv, No. 36791FP).

<sup>30</sup> Mykhailo Krawtchouk and Volodymyr Mozhar, *Diferencialni Rivnyanya ta ich Zastosuvannia v Pryrodoznavstvi i Technitsi* (Kyiv, Vydavnytstvo VUAN, 1934); M. A. Martynenko and V. Y. Vasyanovych, 'Tvorche nadbannya ta ukorochene zhyttya professora Volodymyra Ivanovycha Mozhara,' in *Suchasni Naukovo-Metodychni Problemy Matematyky u Vyschij Schkoli. Vseukrainska Naukovo-Metodychna Konferentsiya* (Kyiv, UDUCHT, 2001).

<sup>31</sup> Mykhailo Krawtchouk, 'O Rabotach Instituta Matematiki Akademii Nauk Ukr. SSR,' *Uspekhi Matematicheskikh Nauk*, 3, 1937, pp. 249-251.

<sup>32</sup> Mozhar and Dubinsky Archival Investigative Case, Vol. 1, pp. 23-33.

<sup>33</sup> *Ibid.*, p. 34.

<sup>34</sup> *Ibid.*, p. 65-82.

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<sup>35</sup> *Ibid.*, p. 141.

<sup>36</sup> *Ibid.*, p. 100.

<sup>37</sup> *Ibid.*, p. 69.

<sup>38</sup> *Ibid.*, pp. 84-88.

<sup>39</sup> Topolyansky Archival Investigative Case (TsGAOO, Kyiv, No. 16986FP), p. 41.

<sup>40</sup> *Ibid.*, pp. 33-38.

<sup>41</sup> *Ibid.*, p. 41.

<sup>42</sup> *Ibid.*, pp. 43-50.

<sup>43</sup> Shapoval, *Ukraine 20-50-kh Rokiv*, pp. 197-201.

<sup>44</sup> After his release, Topolyansky taught at Dnipropetrovsk University.

<sup>45</sup> Krawtchouk Archival Investigative Case, Vol. 2, pp. 25-124.

<sup>46</sup> Tchaikovsky was released in 1943 after serving a 10-year sentence. He became a professor of mathematics at Lviv University.

<sup>47</sup> Lagunov and others Archival Investigative Case (TsGAOO, Kyiv, No. 57955FP).

<sup>48</sup> Archival materials about interrogation of Mykhailo Killerog, a top party official and the former professor at the National Economic University, do not provide sufficient information to determine whether his confession was voluntary or involuntary. See Killerog Archival Investigative Case (SBU Archive, Kyiv, No 33260FP).

<sup>49</sup> See Ivanushkin Archival Investigative Case (SBU Archive, Kyiv, No. 46299FP); Krister Archival Investigative Case (SBU Archive, Kyiv, No. 49819FP); Levytsky Archival Investigative Case (TsGAOO, Kyiv, No. 40541FP); Lukovsky Archival Investigative Case (TsGAOO, Kyiv, No. 38408FP); Polonsky Archival Investigative Case (SBU Archive,

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- Kyiv, No. 69189FP); Ptukha Archival Investigative Case (SBU Archive, Kyiv, No. 39676FP); Suprunenko Archival Investigative Case (SBU Archive, Kyiv, No. 51331FP).
- <sup>50</sup> Feshchenko's Archival Investigative Case (TsGAOO, Kyiv, No. 33272FP).
- <sup>51</sup> *Ibid.*
- <sup>52</sup> Killerog Archival Investigative Case, pp. 42-78.
- <sup>53</sup> See Lagunov and others Archival Investigative Case, Vol. 5, pp. 156-166.
- <sup>54</sup> Volodymyr Prystaiko and Yuri Shapoval, *Mykhailo Hrushevskiy i HPU-NKVD: Trahichne Desiatylittia, 1924-1934* (Kyiv, Ukraina, 1996), pp. 337-338, and Verba, 'Kost Shteppa,' p. 101.
- <sup>55</sup> Shteppa Archival Investigative Case (SBU Archive, Kyiv, No. 49863FP), Vol. 2, p. 181, quoted in Prystaiko and Shapoval, *Mykhailo Hrushevskiy i HPU-NKVD*, p. 347.
- <sup>56</sup> See Verba, 'Kost Shteppa.'
- <sup>57</sup> Volodymyr Kosyk, *Ukraina I Nimmechyna u Drugij Svitovij Vijni* (Lviv, Naukove Tovaristvo Imeni T. Shevchenka, 1993), pp. 227-228.
- <sup>58</sup> See Alexander Dallin, 'Foreword,' in Konstantin Shteppa, *Russian Historians and the Soviet State* (New Brunswick, Rutgers University Press, 1962), Popov, *XX Vek--Istoriia Odnoi Semji* and Verba, 'Kost Shteppa.'
- <sup>59</sup> Beck and Godin, *Russian Purge and the Extraction of Confession* and Shteppa, 'Ezhovshina.'
- <sup>60</sup> Shteppa, 'Ezhovshina,' p. 36.
- <sup>61</sup> See Popov, *XX Vek--Istoriia Odnoi Semji*.
- <sup>62</sup> Volodymyr Prystaiko and Yuri Shapoval, *Sprava "Spilky Vyzvolennia Ukrainy": Nevidomi dokumenty i fakty* (Kyiv, Intel, 1995), p. 372.



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<sup>63</sup> On April 19, 1930, Hermaize was sentenced to 8 years. The NKVD arrested him again in 1937 soon after his release. Hermaize did not provide voluntary confession in this case. Troika, an extrajudiciary body created at the start of the Great Terror, in Saratov region of Russia sentenced Hermaize to 10 years in December 1937.

<sup>64</sup> Mykola Soroka, 'Osoba Sprava Studenta Koroleva,' *Znannya I Pratsya*, 1, 1986, pp. 15-17.

<sup>65</sup> James Harford, *Korolev: How One Man Masterminded The Soviet Drive To Beat America To The Moon* (New York, Wiley, 1997), p. 49 and Yaroslav Golovanov, *Korolev: Fakty I Mify* (Moskva, Nauka, 1994).

<sup>66</sup> Korolev was sent to the same GULAG camp in 1939 where Krawtchouk was sent. However, after others intervened on his behalf, Korolev was transferred from a Kolyma camp to *sharazka*, a blend of a prison and a research institute, in 1940 to work on military projects, and released in 1944.

<sup>67</sup> Two military commanders arrested in Tukhachevsky case provided voluntary confessions and appealed for leniency in exchange for their cooperation with the prosecution. However, five commanders did not confess voluntarily. The behavior of one person during interrogation and trial cannot be determined. All eight commanders were executed.

<sup>68</sup> See Harford, *Korolev*, pp. 49-50.

<sup>69</sup> *Ibid.*

<sup>70</sup> *Ibid.*

<sup>71</sup> Soroka, *Kolymaska Theorema Kravchuka*; Anatoly Datsenko, Yuri Kondratyuk-Oleksander Shargei)-*Syn Ukrainy, Poltava* (Poltava, Poltava, 2000).

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<sup>72</sup> *Ibid.*

<sup>73</sup> Krawtchouk Archival Investigative Case, Vol. 2, pp. 1-7.

<sup>74</sup> Pacharevsky, Petrushevsky, Lysenko, Kondratyuk and Manzhos Archival Investigative Case (TsGAOO, Kyiv, No. 48192FP).

<sup>75</sup> Kondratyuk Archival Investigative Case (TsGAOO, Kyiv, No. 4528FP).

<sup>76</sup> James R. Hansen, *Enchanted Rendezvous: John C. Houbolt and the Genesis of the Lunar-Orbit Rendezvous Concept* (Washington, D.C, NASA, 1995); John M. Logsdon, 'The Choice of the Lunar Orbital Rendezvous Mode,' *Aerospace Historian* 18, 1971, pp. 63-70.

<sup>77</sup> A. V. Datsenko and V. I. Prystchepa, *Yuri Vasilevich Kondratyuk* (Moskva, Nauka, 1997).

<sup>78</sup> See Asif A. Siddiqi, *Challenge to Apollo: The Soviet Union and the Space Race, 1945-1974* (Washington, D.C., NASA, 2000).

<sup>79</sup> Goldman Archival Investigative Case (TsGAOO, Kyiv, No. 33102FP); Kiyak and Proskura, 'Dolya Akademika Oleksandera Goldmana.'

<sup>80</sup> See Yu. V. Pavlenko, Yu. N. Raniuk and Yu. A. Khramov, "*Delo*" *UFTI 1935-1938* (Kyiv, Feniks, 1998), pp. 312-321. In 1941-1944, Leipunsky headed the combined Institute of Mathematics and Physics of the VUAN.

<sup>81</sup> Pavlenko, Raniuk and Khramov, "*Delo*" *UFTI 1935-1938*, pp. 233-244, 282.

<sup>82</sup> See Pavlenko, Raniuk and Khramov, "*Delo*" *UFTI 1935-1938* and Weissberg, *The Accused*.